Melbourne Freeway Performance & Congestion Monitoring Dashboard

Operational reporting brief for corridor performance, travel time, delay, and congestion monitoring using Victorian transport data.

Project Sponsor / Function	Transport Operations / Network Performance Reporting
Reporting Product	Power BI operational monitoring dashboard
Geographic Focus	Metropolitan Melbourne freeway network

1. Business Scenario

A transport operations team requires a consistent reporting view of freeway conditions across Melbourne. While traffic performance data is available, operational stakeholders still need a structured BI layer that converts raw measures into a clear view of corridor performance, delay, congestion, and route-level pressure.

The proposed dashboard will provide a repeatable monitoring product that supports operational review, trend assessment, and the identification of locations that warrant closer attention.

2. Business Problem

Freeway performance data is high-frequency and difficult to interpret directly in raw form. Without a structured dashboard, analysts and managers can struggle to identify where delays are emerging, which corridors are under the most pressure, when conditions deteriorate, and which segments show the greatest operational concern.

The core problem is not data availability; it is turning raw transport data into trusted, repeatable, decision-useful reporting.

3. Project Objective

Design and build a Power BI dashboard that monitors freeway travel time performance and congestion conditions across Melbourne using Victorian transport data.

The dashboard should support corridor-level performance review, time-based trend analysis, delay monitoring, and exception identification through a clean and usable reporting interface.

4. Stakeholder Audience

- Transport Operations Analysts responsible for monitoring freeway conditions and operational pressure points.
- Network Performance or Reporting Analysts supporting regular performance review and management reporting.
- Corridor planning or performance teams reviewing recurring congestion patterns and route underperformance.
- Operational managers who require an accessible summary of network conditions and emerging issues.

5. Scope

In Scope	Out of Scope
• Travel time, speed, congestion, delay, and related traffic-condition indicators.	• Production streaming architecture or complex engineering pipelines.
• Corridor, route, direction, and segment comparisons where supported by the data model.	• Machine learning or predictive forecasting in the initial release.
• Time-based analysis such as hourly patterns, weekday trends, and recurring congestion windows.	• Broad integration of external datasets that add complexity without clear reporting value.
• A KPI-led Power BI report designed as an operational monitoring product.	• A map-first exploratory product that prioritises visual novelty over operational usefulness.
• Practical metric definitions, caveats, and data checks.	

6. Key Business Questions

- Which freeway corridors and directions show the highest levels of congestion and delay?
- At what times do travel conditions deteriorate most significantly?
- Which segments appear to underperform most consistently relative to nominal travel conditions?
- Where do the most operationally significant delay hotspots appear across the monitored network?
- What patterns should stakeholders prioritise for monitoring, review, or further investigation?

7. Data Source Direction

The proposed reporting product will use the Victorian Freeway Travel Time data collection as the core source. This collection includes traffic-condition measures such as travel time, speed, congestion, and delay, supported by GIS/location metadata for route, segment, and directional context.

Initial delivery should use a controlled extract or snapshot approach rather than a live streaming design, keeping the solution practical, stable, and suitable for analytical reporting.

8. Delivery Approach

- Use the traffic dataset as the core fact source and supporting GIS data only where it improves route, segment, or directional interpretation.
- Limit the initial release to a focused set of high-value KPIs, corridor comparisons, and time-based trend views.
- Document data limitations clearly so the reporting product remains credible and interpretable.
- Keep the model and report structure simple enough to support maintenance, review, and clear stakeholder use.

9. Deliverables

- Approved project brief and scope documentation.
- Documented data dictionary and practical data checks.

- Transformation and modelling notes covering Power Query, data preparation, and core measures.
- Power BI report with a clean multi-page operational dashboard structure.
- Findings summary and recommendations for operational review.

10. Success Criteria

- The dashboard reads as a credible operational reporting product with clear monitoring value.
- KPI definitions are useful, transparent, and aligned to stakeholder questions.
- The data model and visuals remain clear, consistent, and easy to interpret.
- The reporting output supports corridor comparison, time-based review, and issue identification without unnecessary complexity.

11. Risks and Control Points

- **Risk: Scope drift into complex engineering or geospatial work.** Control: Keep the build focused on analyst-grade reporting outputs and controlled data extracts.
- **Risk: Too many visuals or overlapping measures.** Control: Prioritise a compact KPI set and a small number of decision-useful report pages.
- **Risk: Weak metric interpretation due to data limitations.** Control: Define caveats, field meanings, and quality checks early in the project.
- **Risk: Reduced usability from overdesigned layout choices.** Control: Use a simple operational design standard with clear hierarchy and restrained styling.